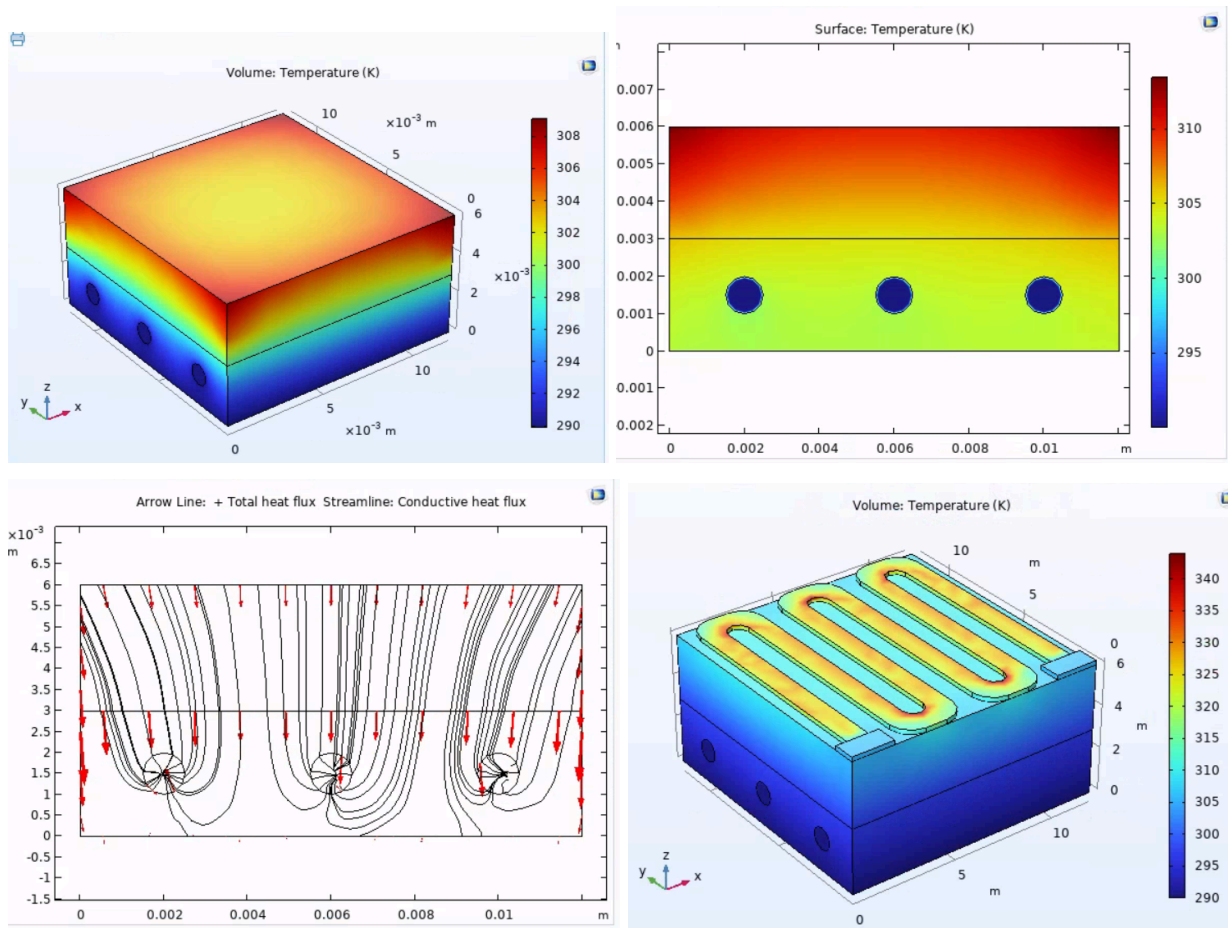
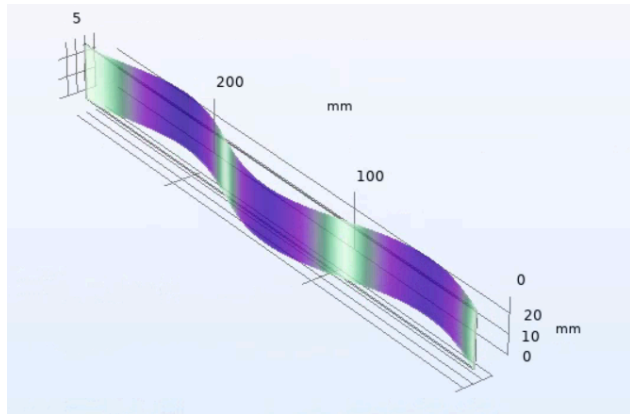
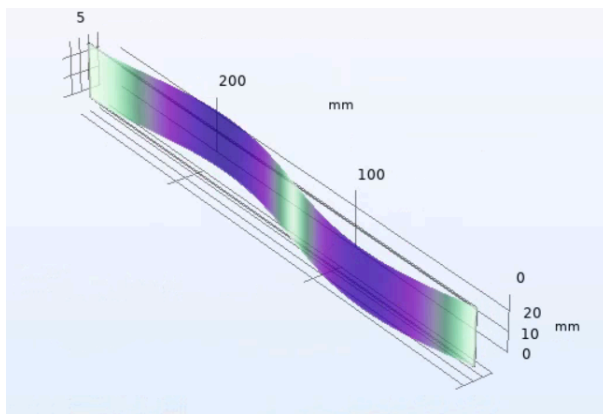
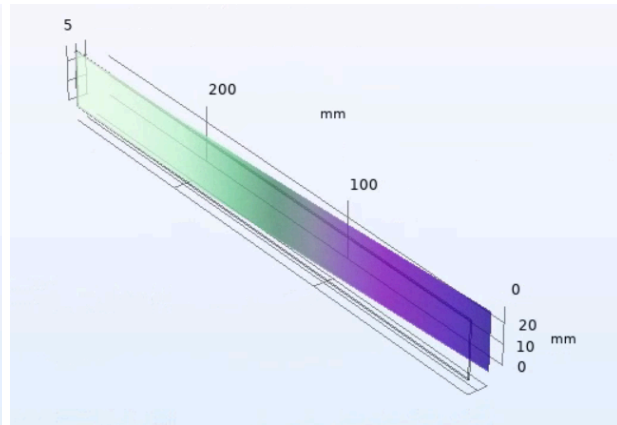
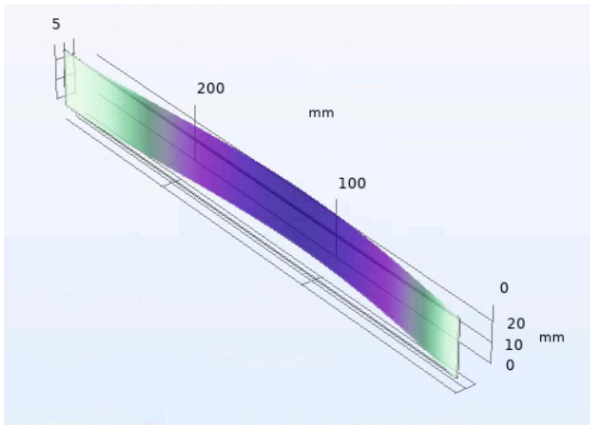


Miscellaneous Projects



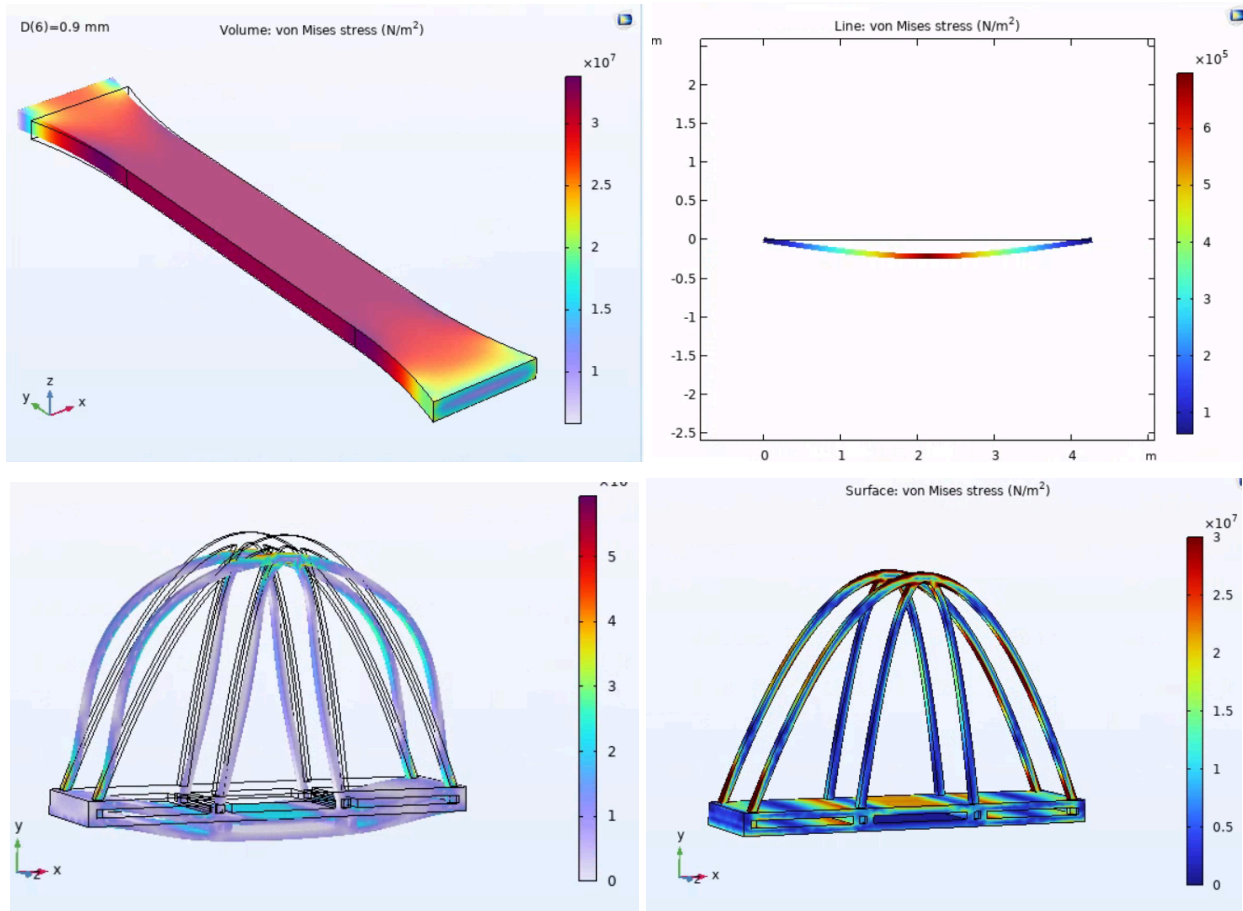
ES183: Heat Transfer

ES183 was one of my first introductions to FEA and COMSOL. This assignment was basically a sort of trial-by-fire tutorial on how to use the software. We were tasked to impose different boundary conditions such as radiative heat flux vs. electric coil heating, constant temperature heat sink vs. insulated bottom, all with liquid coolant flowing at constant temperature to compare and contrast the way heat was distributed at steady state across differing but similar circumstances.



ES125: Mechanical Systems

This assignment was just a basic modal analysis of different harmonic vibration modes of a beam. In class we had abstracted ideas of masses and springs to continuous systems described by wave equations. This project was meant to give a visual and numerical aid to help in understanding the eigen modes of a cantilevered beam, in this case.



ES120: Solid Mechanics

ES120 had various assignments throughout the semester for students to learn COMSOL and study stress, strain, yield, etc. In the upper left we have a model prescribed some boundary force conditions and some Young's modulus. We actually took different materials formed into this shape and tested them in an Instron machine to compare to the simulation. On the bottom we have the final class project – a bridge building competition. My group and I tested different designs in COMSOL before choosing the one with the best strength to weight ratio to build and compete.